

SERVICE - ORIENTED SOFTWARE ENGINEERING

DT0203

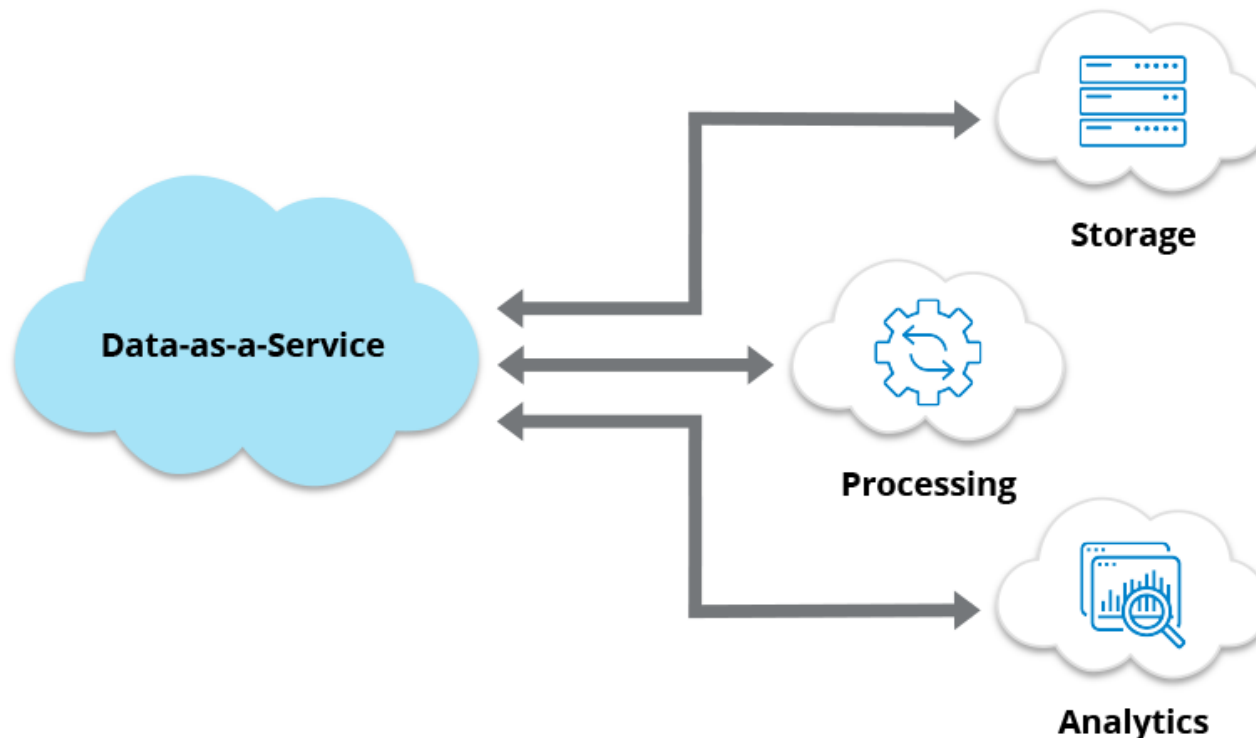
Data as a Service (DaaS)

Prof. Marco Autili
University of L'Aquila
marco.autili@univaq.it

What is a Data as a Service?

Data as a Service (DaaS)

- data management “strategy” that focuses on public or private cloud to deliver a variety of data-related services, such as storage, processing, and analytics



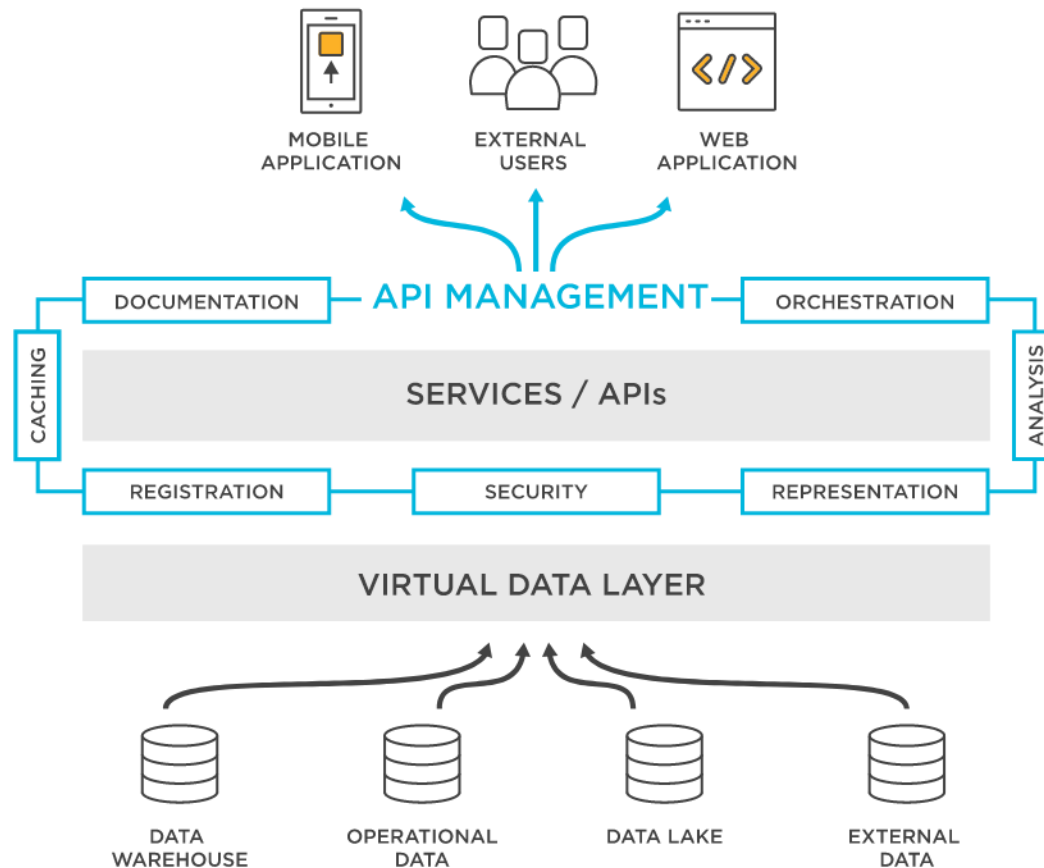
What is a Data as a Service?

Data as a Service (DaaS)

- aims to leverage **data** as a **business asset** for greater business agility
- uses a **cloud-based underlying technology** that supports **Web Services** and **SOA**
- leverages the popular **Software-as-a-Service (SaaS)** paradigm
 - **cloud computing strategy** that involves delivering applications to end-users over the network (regardless of geographic or organizational separation between provider and consumer.)
- **outsources** most data storage, integration, and processing **operations to the cloud**
- provides a way to **manage massive amounts of data** to **deliver valuable information and insights** for **data-driven decision making**. This is **critical** for any **business wanting to turn data into real value**

What is a Data as a Service?

- DaaS focuses on provisioning data from a variety of sources on demand through APIs
- DaaS delivers datasets or streams of data to be consumed in a range of formats, often unified using data virtualization



Benefits of Data as a Service

- **Monetizing Data:** having enough data is no longer a major issue, organizing and operationalizing the data is the biggest challenge. DaaS could be a key way to leverage the full value of data
- **Lower costs:** exploit all the wide range of business data sources to uncover insights and delivering them to different areas of the business allows to act smarter and to greatly decrease spending time and money on the wrong decisions. DaaS means wasting less resources on pointless efforts and taking more data-driven decisions
- **Faster Paths to Innovation:** DaaS opens the doors to growth. With data at the center of a business, growth happens quickly. Data-informed strategies allow for more innovation with less risk.

Benefits of Data as a Service

- **More Agile Decision Making:** DaaS allows businesses to treat data as an important business asset for more strategic decision making and effective data management. It can combine both internal and external data sources (customer, partner, and open data sources) for a comprehensive view of the business
- **Data-Driven Culture:** DaaS allows businesses to deliver integrated data from a growing list of data sources, fostering a data-driven culture and democratizing the use of data in everyday processes
- **Lower Risks:** DaaS helps lower risks associated with conflicting or incomplete data views or poor data quality. Companies can leverage data virtualization and other technologies to access, combine, transform, and deliver data via reusable data services, optimizing query performance and ensuring data security and governance

Criticisms of Data as a Service

- **Privacy**: when an organization chooses to share and deploy data outside their own organization, privacy becomes a main concern. In a DaaS environment, privacy concerns are significant because the data that is being shared is deep and often tied to mission-critical applications that directly affect key business processes
- **Security**: from a security perspective the easy access of data in a DaaS environment can potentially leave business-critical datasets open to a variety of vulnerabilities. As more and more organizations move to adopt DaaS as a viable option, robust security plans will need to be put in place to avoid major security breaches
- **Data Governance**: must rely on more than the nature of a Cloud platform to ensure high levels of Data Governance. The integrity of data must be tested and checked to ensure that it harmonizes with any other data. Verification on this level can be tough to implement

Challenges of Data as a Service

- **Security:** DaaS requires organizations to move data into cloud infrastructure and transfer data over the network, this can create security risks that have to be mitigated
- **Additional compliance steps:** when sensitive data is moved into a cloud environment compliance challenges may also arise. Companies subject to special data compliance requirements must ensure that they meet those requirements with their DaaS solution (for example, host their DaaS on cloud servers located in a specific country)
- **Potentially limited capabilities:** users are able to work only with the tools that are hosted on or compatible with a DaaS platform, rather than being able to use any tools of their choice to set up their own data-processing solutions. Choosing a DaaS solution that offers maximum flexibility in choosing tools mitigates this challenge

Challenges of Data as a Service

- **Data transfer timing:** transferring large volumes of data into a DaaS platform can take time due to network bandwidth limitations. This can pose a serious challenge if an organization needs to frequently move data into a DaaS platform. Data compression and edge computing strategies can help to accelerate transfer speeds
- **Data hygiene:** when an organization works with a DaaS vendor, their internal data may have to be combined with the provider's data. Vendors' and the organization's rules for data preparation may not match and this leads to dirty data. Organizations should ensure that the vendor understands how to cleanly sync with their data

Key elements of Data as a Service

- DaaS provides a data analytical platform using a cloud-based solution
 - multiple key elements are involved in DaaS to draw out actionable insights from data
-
1. **Data collection:** process of identifying the best methodology for gathering structured and unstructured data from various data sources. Once the data is collected, it is stored and loaded into data warehouses on the cloud. This step ensures the integrity of the data collected and it helps to clarify if the data collected is correct while reducing the probability of errors

Key elements of Data as a Service

2. **Data aggregation**: assembling and presenting the data in summarized manner. The structured and unstructured data collected is combined and merged into a summarized metadata for analysis. The most important process is to identify the accuracy of hidden patterns from the data analysis which depends upon the volume, integrity and quality of the data collected. It is crucial to collect high-volume and high-quality accurate data to get the expected results. Data aggregation is helpful for finance, business decisions, market planning, operations, and pricing
3. **Data correlation**: the process of using statistical analysis to find a strong relationship between two or more data points and to determine how the data is associated and mapped. A strong correlation implies a strong relationship between two data points, improving the data analysis and decision-making processes

Key elements of Data as a Service

4. **Data visualization**: process of identifying patterns and hidden insights by creating a visual representation for any organization, stakeholders, or end user. The data is analyzed and represented in interactive and dynamic dashboards, charts, graphs, etc. These visualizations give business teams the ability to formulate marketing, sales, finance and management strategies
5. **Data analytics**: process of building complex models around the data to find hidden patterns and gain insights from large and complex data. This analysis simplifies big data problems, unlocking new insights and avoiding analysis paralysis

References

- <https://www.tibco.com/reference-center/what-is-daas>
- <https://www.rocketsource.co/blog/data-as-a-service/>